

Daily Combinatorics Drill #2

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Combinations $\binom{n}{r}$ · Choosing vs Arranging · Complementary Counting · Committee Constructions.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Evaluate or count instantly. Choosing, not arranging. Pure recognition.

1. Evaluate: $\binom{7}{2}$.
2. Evaluate: $\binom{8}{3}$.
3. Evaluate: $\binom{9}{7}$.
4. How many committees of 3 can be chosen from 6 people?
5. Solve for n : $\binom{n}{2} = 45$.
6. How many subsets of a 5-element set have at least 4 elements?
7. Evaluate: $\binom{6}{0} + \binom{6}{1} + \binom{6}{2} + \cdots + \binom{6}{6}$.
8. How many ways can 2 letters be chosen from {A, B, C, D, E}?
9. True or false: $\binom{100}{98} = 4950$.
10. A lottery ticket selects 4 numbers from 1–10 (order irrelevant). How many different tickets are there?

Section B Manipulation Drills

(≤50 s each)

Set up the binomial coefficient, don't list. Watch for choose-then-choose products.

1. Solve for n : $\binom{n}{3} = 35$.

2. From 8 men and 6 women, how many committees of 2 men and 2 women can be formed?
3. Twelve people at a party all shake hands with each other exactly once. How many handshakes take place?
4. Ten points lie in the plane, no three collinear. How many line segments join pairs of them, and how many triangles have all three vertices among them?
5. A team of 5 is chosen from a squad of 11, and then one of the 5 is named captain. How many ways? Check your answer by choosing the captain first.
6. How many binary strings of length 8 contain exactly three 1s?
7. From the 5 vowels and 21 consonants, how many 3-letter *sets* consist of exactly one vowel and two consonants?
8. Solve for n : $\binom{n}{2} + \binom{n}{1} = 21$.
9. A committee of 4 is chosen from 9 people. How many committees include Priya? How many exclude her?
10. A robot walks from $(0,0)$ to $(4,3)$ taking unit steps only right or up. How many routes are there?

Section C Substitution & Structure

(≤90 s each)

Case splits and complements. Decide "at least / at most / exactly" before computing.

1. A committee of 5 is chosen from 7 women and 6 men and must contain at least 4 women. How many committees are possible?
2. How many four-digit numbers have at least one repeated digit?
3. How many diagonals does a regular 12-gon have?
4. Eight points lie on a circle. How many triangles have all vertices among them? How many of those triangles *avoid* the point A ?
5. From 6 boys and 6 girls, a team of 6 is chosen with *unequal* numbers of boys and girls. How many teams?
6. How many 5-card hands from a standard 52-card deck contain at least three aces?
7. A convex n -gon has exactly 27 diagonals. Find n .
8. How many subsets of $\{1, 2, \dots, 10\}$ contain at least one of the elements 1 and 2?

Section D Challenge

(SMC difficulty)

Insight over computation. If the last ones resist, sit with them — learn the tool, then return.

1. How many rectangles (of any size) can be drawn on a 4×4 grid of unit squares with sides along the gridlines? (*extends Sheet 1, D2*)
2. How many subsets of $\{1, 2, \dots, 8\}$ contain no two consecutive integers? (Count the empty set.) (*cf. MAT 2023 Q5*)
3. A club has 6 juniors and 6 seniors. How many committees (of any size) contain strictly more juniors than seniors?
4. Six people each shake hands with exactly two of the others. In how many ways can this happen?
5. Ten points lie on a circle, and every pair is joined by a chord. If no three chords meet at a single interior point, how many interior intersection points are there?

“When a count splits into cases, count the complement first — the opposite of ‘at least one’ is always simpler.”

other prep to look at today:
Daily Integral for Calculus and Logic